Module Name: Browser Navigation, Form Handling, Web Element Interaction and Validation - Python Selenium Automation Testing

Test Case ID: st_14

Tester Name: Aman Sevak

Webpage URL: https://www.saucedemo.com/

Tools Used: Python, Selenium WebDriver, PyCharm, Google Chrome

- ## Scenario ID: 1a

Scenario: Open the SauceDemo website in Chrome

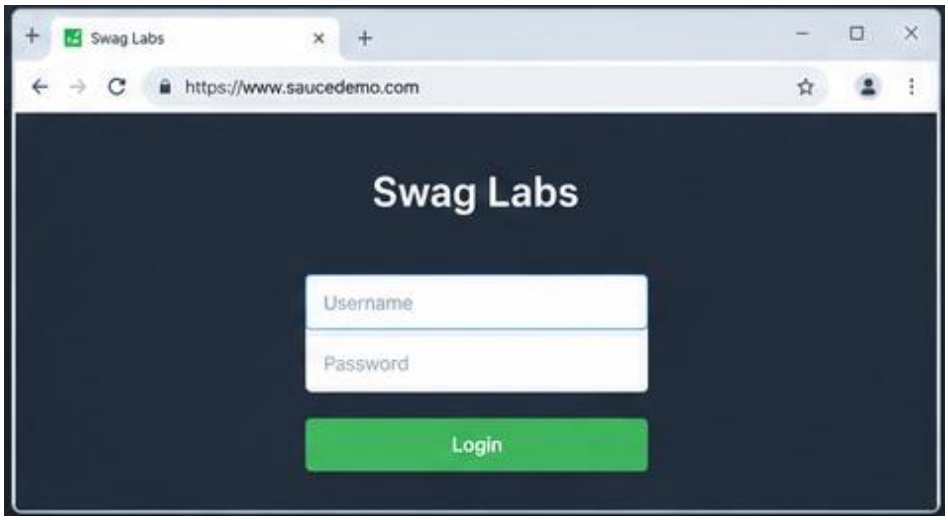
Test Steps:

1. Launch Google Chrome using Selenium WebDrivers
2. Maximize the browser window
3. Open https://www.saucedemo.com/
4. Print the page title

Expected / Actual Result: Website opened successfully. Login page is displayed.

Status: **Pass**

Screenshots:



- ## Scenario ID: 1b

Scenario: Login with valid credentials

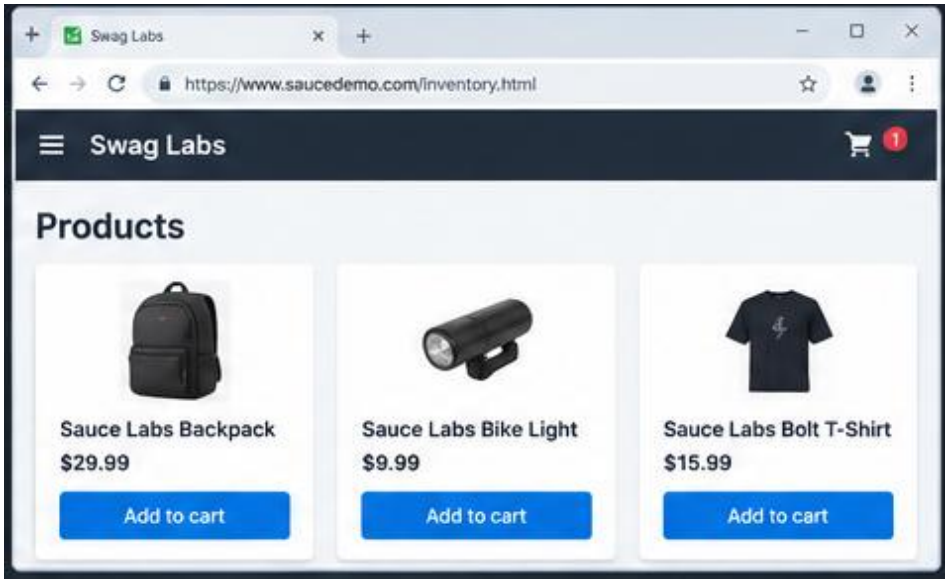
Test Steps:

1. Enter Username: standard_user
2. Enter Password: secret_sauce
3. Click on Login
4. Validate that the URL contains "inventory"

Expected / Actual Result: User logged in and Products page (inventory) opened.

Status: Pass

Screenshots:



- ## Scenario ID: 1c

Scenario: Select a product (Add to Cart)

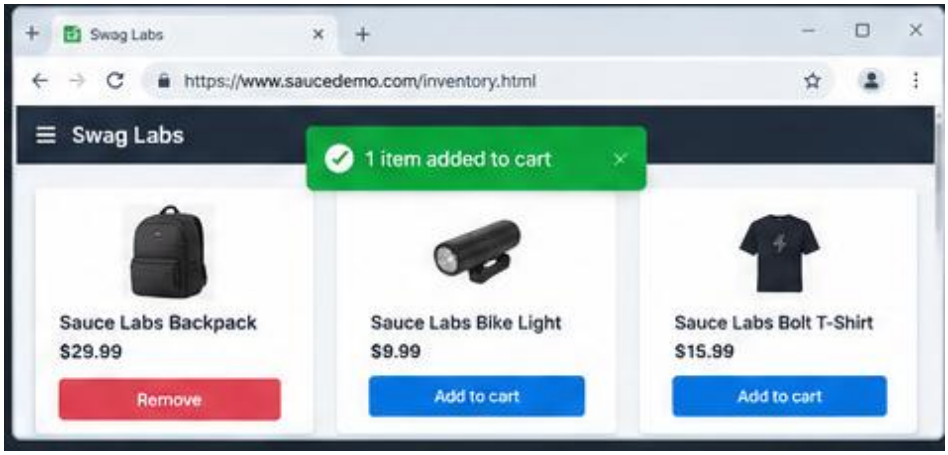
Test Steps:

1. Wait until "Add to cart" button of Sauce Labs Backpack is clickable
2. Click on Add to cart

Expected / Actual Result: Product added to the cart; cart badge updated.

Status: Pass

Screenshots:



- ## Scenario ID: 1d

Scenario: Cart navigation

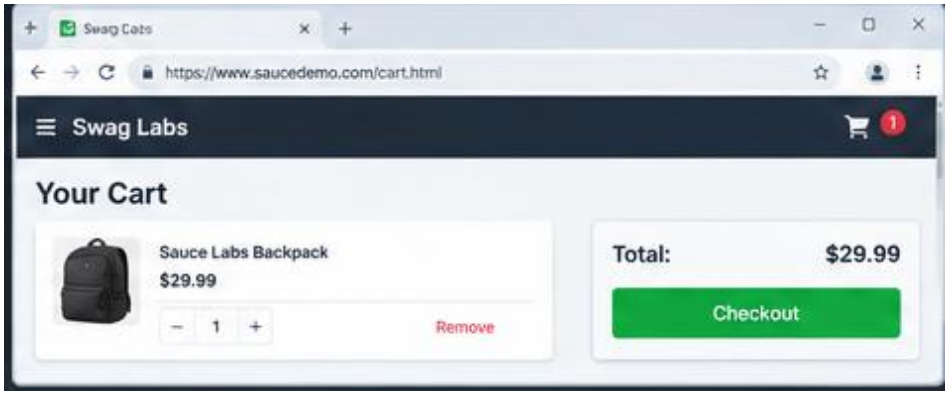
Test Steps:

1. Click on the shopping cart icon
2. Validate that the URL contains "cart"

Expected / Actual Result: Cart page opened and selected product is listed.

Status: Pass

Screenshots:



- **Source Code (tutorial14.py)**

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

driver = webdriver.Chrome()
driver.maximize_window()

wait = WebDriverWait(driver, 10)

try:

    # Browser Navigation
    driver.get("https://www.saucedemo.com/")

    print("Page Title:", driver.title)

    # Form Handling
    wait.until(
        EC.visibility_of_element_located(
            (By.ID, "user-name")
        )
    ).send_keys("standard_user")

    driver.find_element(
        By.ID, "password"
    ).send_keys("secret_sauce")

    # Web Element Interaction
    driver.find_element(
        By.ID, "login-button"
    ).click()

    # Validation
    wait.until(
        EC.url_contains("inventory")
    )
    assert "inventory" in driver.current_url

    print("Login Validation: PASS")

    # Product Interaction
    wait.until(
        EC.element_to_be_clickable(
            (By.ID, "add-to-cart-sauce-labs-backpack")
        )
    ).click()
```

```
    print("Product Selection: PASS")

    # Cart Navigation
    driver.find_element(
        By.CLASS_NAME,
        "shopping_cart_link"
    ).click()

    assert "cart" in driver.current_url

    print("Cart Navigation: PASS")
    print("All Selenium Web Automation Tests Passed")

finally:
    driver.quit()
```

- ## Console Output (PyCharm)

```
Page Title: Swag Labs
Login Validation: PASS
Product Selection: PASS
Cart Navigation: PASS
All Selenium Web Automation Tests Passed
```

- ## Summary Of Test Results:

Total Test Cases- 4

Passed- 4

Failed- 0

Pass %- 100%

Conclusion and Recommendations:

Pass Rate: 100%, All Tests Passed

The Selenium WebDriver script successfully automated website navigation, login form handling, product selection and cart navigation of the SauceDemo application. All assertions passed and the URL validations for the inventory and cart pages worked as expected.

Issues Found

No issues or defects were found during execution of the automated test scenarios.

Recommendations

1. **Add Negative Test Cases**
   Extend the script with invalid login (locked_out_user, wrong password) and empty login field validation scenarios.
2. **Use Page Object Model**
   Separate locators and actions into page classes to make the script easier to maintain.
3. **Run as Regression Suite**
   Re-execute the script after every change to confirm that existing functionality is unaffected.

- **Bug Reports:**

No bugs were reported for this practical, as all test cases passed.